AI Fundamentals — Bootcamp Module 1

Deep Learning & Neural Networks

What Is a Neural Network?

A neural network is a layered mathematical function loosely inspired by neurons in the brain. It consists of nodes (neurons) organised in layers, connected by weighted edges. The network learns by adjusting those weights.

Three-Layer Architecture

Input Layer: Receives raw data — pixel values, word tokens, sensor readings.

Hidden Layers: Transform inputs through weighted sums and non-linear activation functions. 'Deep' = many hidden layers.

Output Layer: Produces the prediction — a class label, a number, a probability distribution.

> *'Deep Learning' simply means a neural network with many hidden layers — 'deep' refers to this depth, not mystery.*

How Learning Happens — Backpropagation

Backpropagation is the error-correction mechanism that trains neural networks:

- 1. Forward pass: data flows through the network, producing a prediction.
- 2. Loss calculation: compare prediction to the true label using a loss function (e.g., cross-entropy).
- 3. Backward pass: compute the gradient of the loss with respect to each weight (chain rule calculus).
- 4. Gradient descent: nudge each weight slightly in the direction that reduces loss.
- 5. Repeat millions of times across the dataset until loss is minimised.

> *Analogy: Backprop is like getting a test back with marks — you see where you went wrong and adjust your study (weights) accordingly.*

Activation Functions

ReLU (Rectified Linear Unit): Outputs 0 for negative inputs, passes positive values unchanged. Most common in hidden layers.

Sigmoid: Squashes output to 0–1. Used in binary classification output layers.

Softmax: Converts a vector of numbers to a probability distribution. Used for multi-class output.

Types of Neural Networks

CNN (Convolutional Neural Network): Specialised for grid data — images and video. Uses convolutional filters to detect edges, textures, objects.

RNN (Recurrent Neural Network): Designed for sequences — text, speech, time series. Has 'memory' of previous steps.

Transformer: State-of-the-art for language and multi-modal tasks. Uses self-attention instead of recurrence. Powers GPT, BERT, Claude.

Key Takeaways

- Neural networks are layered mathematical functions: Input → Hidden → Output.
- Learning occurs via backpropagation: forward pass → loss → backward pass → weight update.
- Activation functions introduce non-linearity, enabling the network to learn complex patterns.
- CNNs excel at images; RNNs at sequences; Transformers at language and beyond.

Quick-Revision Questions

- Describe the three layers of a neural network and each layer's role.
- Explain backpropagation in four steps.
- Why are Transformers now preferred over RNNs for language tasks?